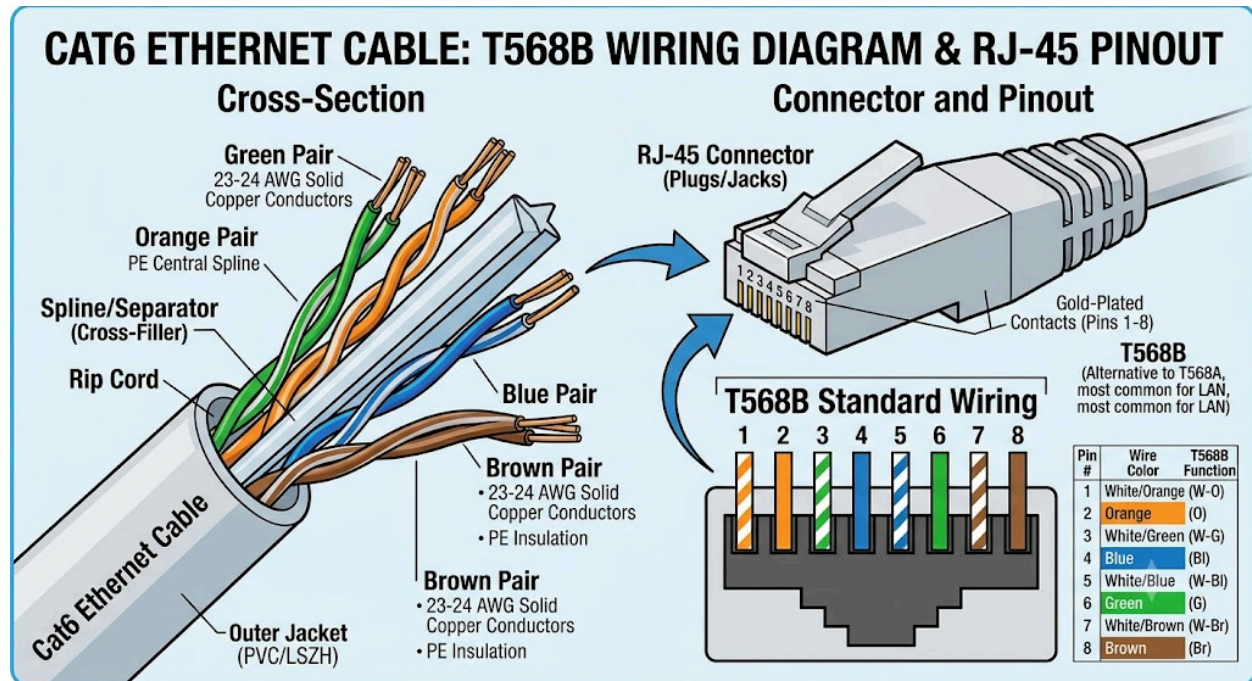


NETWORK CABLE AND CONNECTORS

Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

ANS:



List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

ANS:

Cat5e Cable: Used to connect a **Zomato POS terminal** or desktop computer to a local network switch. It is inexpensive and provides reliable 1 Gbps speed over short distances, which is more than sufficient for handling lightweight transaction data.

Cat6 Cable: Used to connect a high-speed **Wi-Fi router** to a core network switch in an office or home. Its internal plastic spline and tighter twists reduce crosstalk, allowing it to reliably transmit up to 10 Gbps speeds over shorter distances (up to 55 meters).

Fiber Optic Cable: Used to link high-bandwidth **IPL streaming servers** directly to cloud distribution nodes or ISP backbones. It transmits data using light signals across immense distances with zero electromagnetic interference, delivering ultra-high bandwidth and sub-millisecond latency needed for mass video delivery.

Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:

Step-by-Step Process for Crimping an RJ-45 Connector

1. **Strip the Cable Jacket:** Use a cable stripper to carefully remove roughly 1 inch (2.5 cm) of the outer PVC jacket without nicking the inner copper wires.
2. **Trim the Plastic Spline (if using Cat6):** Cut away the central plastic separator/spline at the base of the exposed wires using flush cutters.
3. **Untwist & Straighten Wires:** Untwist the four wire pairs all the way to the jacket edge and flatten them out straight using your fingers.
4. **Arrange Wires in T568B Order:** Align the wires side-by-side from left to right according to the standard:
 - **1:** White/Orange
 - **2:** Orange
 - **3:** White/Green
 - **4:** Blue
 - **5:** White/Blue
 - **6:** Green
 - **7:** White/Brown
 - **8:** Brown
5. **Trim Wires Flush:** Cut the aligned wires straight across so they extend about half an inch (1.2 cm) from the outer jacket.
6. **Insert into RJ-45 Connector:** Hold the RJ-45 plug with the locking tab facing down and slide the wires firmly into the connector until all copper ends reach the front tip.
7. **Crimp the Connector:** Insert the RJ-45 head into the crimping tool die and squeeze the handles firmly to push the gold pins into the conductors and lock the cable strain relief.

Two Common Mistakes to Avoid

1. **Not pushing the outer jacket inside the plug:** Leaving exposed wires outside the RJ-45 plug causes strain on the internal connections, leading to wire failure over time. The outer jacket must sit under the internal crimp wedge.
2. **Cutting wires unevenly or out of order:** If the wires are cut at an angle or cross over each other inside the plug, pins won't make contact with all conductors, causing connection drops or failure to negotiate Gigabit speeds.

Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

ANS:

For an IPL screening event and multi-console gaming zone, you need separate cable setups for **Video Display** (low input lag and 4K/120Hz refresh rates) and **Local Networking** (zero latency spikes for multiplayer gaming).

1. Video Connections (Consoles to Screens)

- **Cable Type: Ultra High Speed HDMI 2.1 Cables**
- **Connector Type: HDMI Standard Type-A**
- **Why:** HDMI 2.1 supports up to 48 Gbps bandwidth, allowing uncompressed 4K video at 120Hz refresh rates. Features like **ALLM (Auto Low Latency Mode)** and **VRR (Variable Refresh Rate)** eliminate screen tearing and minimize input lag, which is critical for real-time multiplayer gaming and stutter-free live sports broadcasting.

2. Network Connections (Consoles to Network Switch / Router)

- **Cable Type: Cat6 or Cat6a Shielded (S/FTP) Ethernet Cables**
- **Connector Type: Standard RJ-45 Connectors**
- **Why:** Cat6/Cat6a provides up to 10 Gbps speeds with superior shielding against electromagnetic interference (EMI) from surrounding power cables and display equipment. Wired RJ-45 connections ensure stable sub-millisecond local latency, preventing packet loss and ping spikes during competitive gaming matches.

3. Extended Screen Feed (Long Runs to Main Projection Displays)

- **Cable Type: HDMI over Cat6 Extenders (or Active Optical HDMI Cables - AOC)**
- **Connector Type: RJ-45 (for Extenders) / HDMI Type-A**
- **Why:** Standard HDMI cables degrade signal quality after 5 to 7 meters. Using HDMI extenders over Cat6 allows high-definition, low-latency video transmission across lengths up to 70–100 meters to reach elevated stadium-style screens or large projection setups around the venue without video dropouts.

Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

ANS:



- **Cable Type: Cat6 Unshielded Twisted Pair (UTP) Ethernet Cable** (the blue cable shown above)[cite: 3].
- **Connector Type: Clear Plastic RJ-45 Connector** with an internal T568B pin configuration.
- **Devices Connected:** It connects the LAN port on the back of a **Wi-Fi router / fiber ONT gateway** to a desktop computer (or network switch).

Function in Action

This physical connection transmits digital data packets at speeds up to 1 Gbps / 10 Gbps between the router's local switch and the connected device, providing low-latency, interference-free internet access compared to wireless Wi-Fi.